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SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM-2

Programme:B.E-ECE
Course Code: SECB1302
Course Name: Signals and Systems
Batch : 2024 -2028

Max. Marks: 50
Time: 2 Hours
Sem: III
Date: 10.10.2025

Part-A Answer ALL the questions (7 × 2=14)

Q.No	Questions	CO
1.	Define ROC in S-plane	3
2.	Test Whether, $y(t)=x(t^2)$ is linear or not?	3
3.	Check Whether, $y(t)= \cos[x(t)]$ is stable or not?	3
4.	Test whether the following system is time invariance or not? $Y(n)=2n\, x(n)$	4
5.	Define Convolution.	4
6.	Determine the z-transform and their ROC of the given signal , $x(n)=0.3^n\, u(n)$	4
7.	What is impulse response and explain its significance.	4

Part-B Answer ALL the questions (3 × 12 = 36)

8.	Explain the different classification of Discrete time systems in detail (any 5).	3
	(OR)	
9.	State whether the system $y(t)= t\, x(-t)$ is causal, stable, linear and time invariant.	3
10.	Using Laplace Transform , determine the complete response of the system described by the equation. $\frac{d^2 y(t)}{dt^2}+5\frac{dy(t)}{dt}+4y(t)=\frac{dx(t)}{dt};y(0)=0;\frac{dy(t)}{dt}\Big\|_{t=0}=1$ For the input, $x(t) = e^{-2t}\, u(t)$	3
	(OR)	
11.	Explain the properties of Z-transform in detail. (any 6)	4

12.	Determine the response of the LTI system using Matrix method and Graphical method, whose input x(n) and impulse response h(n) are given by, x(n)={1, 2, 0.5, 1} and h(n)={1, 2, 1, -1}	4
	(OR)	
13.	Determine the inverse Z-transform of $X(Z) = \frac{1}{1-4.5Z^{-1}-3.5Z^{-2}}$ by long division method a) if ROC: |Z| > 3.5 b) if ROC: |Z| < 1.0	4